

FitHub AI

“Train smarter. Live better.”

Comprehensive Software Project Life Cycle Document

Project Name	FitHub AI
Application Type	AI-Powered Fitness Coaching Web Application
Primary Tech Stack	Python (FastAPI), Jinja2, SQLAlchemy, SQLite, Google Gemini API, Docker, Uvicorn

8. Project Demonstration Phase

Key Live Workflows Showcased

- Home Page & Metrics Widget:** Displayed hero section with tagline “Train smarter. Live better.”, progress ring visual (74%), step counter (12k steps), and active tracking minutes.
- Interactive Planner Processing:** Demonstrated parameter collection across 13 unique health indicators.
- Dynamic AI Workout Generation:** Demonstrated real-time generation of custom workout routines (e.g., 260 kcal burn plan categorized into Warm-up, Cardio, Legs, and Upper Body).

Future Development Roadmap

- Implement individual user login and session authentication.
- Persist user progress history and historical score trends within the database.
- Implement automated unit testing suites for form inputs and API responses.
- Configure production SSL/HTTPS and link a custom web domain.